

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 // Node structure
5 struct Node
6 {
7     int data;
8     struct Node *next;
9 };
10
11 struct Node *front = NULL;
12 struct Node *rear = NULL;
13
14 // Insert element
15 void enqueue(int value)
16 {
17     struct Node *newNode;
18
19     newNode = (struct Node*)malloc(sizeof(struct Node));
20
21     newNode->data = value;
22     newNode->next = NULL;
23
24     if(rear == NULL)
25     {
26         front = rear = newNode;
27     }
28     else
29     {
```

```
29 {
30     rear->next = newNode;
31     rear = newNode;
32 }
33
34 printf("%d inserted\n", value);
35 }
36
37 // Delete element
38 void dequeue()
39 {
40     struct Node *temp;
41
42     if(front == NULL)
43     {
44         printf("Queue Underflow\n");
45     }
46     else
47     {
48         temp = front;
49
50         printf("%d deleted\n", front->data);
51
52         front = front->next;
53
54         if(front == NULL)
55             rear = NULL;
56
57         free(temp);
```

```
58 }
59 }
60
61 // Display queue
62 void display()
63 {
64     struct Node *temp;
65
66     if(front == NULL)
67     {
68         printf("Queue is Empty\n");
69     }
70     else
71     {
72         temp = front;
73
74         printf("Queue Elements:\n");
75
76         while(temp != NULL)
77         {
78             printf("%d ", temp->data);
79             temp = temp->next;
80         }
81     }
82
83     printf("\n");
84 }
85
86 int main()
```

```
72     temp = front;
73
74     printf("Queue Elements:\n");
75
76     while(temp != NULL)
77     {
78         printf("%d ", temp->data);
79         temp = temp->next;
80     }
81 }
82
83 printf("\n");
84 }
85
86 int main()
87 {
88     enqueue(10);
89     enqueue(20);
90     enqueue(30);
91
92     display();
93
94     dequeue();
95
96     display();
97
98     return 0;
99 }
100
```

10 inserted
20 inserted
30 inserted
Queue Elements:
10 20 30
10 deleted
Queue Elements:
20 30

...Program finished with exit code 0
Press ENTER to exit console.